Banknotes with K-Means

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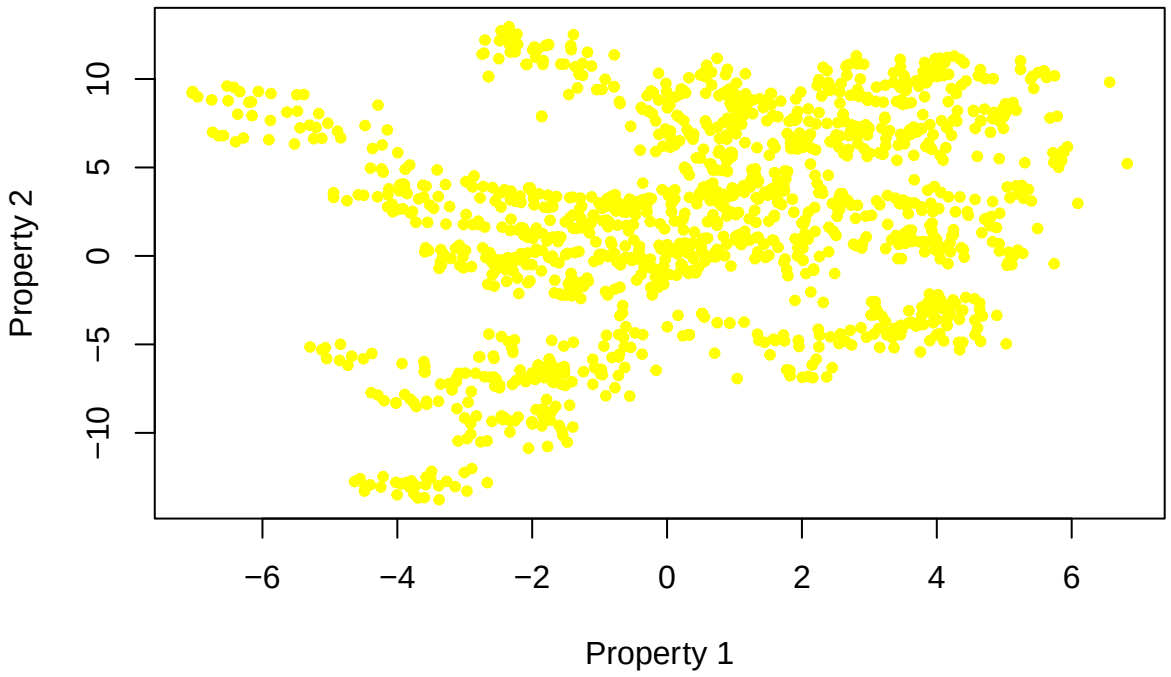
First, we need to import data:

```
notes<-read.csv("banknotes.csv")
head(notes)
```

```
##        V1      V2
## 1 3.62160  8.6661
## 2 4.54590  8.1674
## 3 3.86600 -2.6383
## 4 3.45660  9.5228
## 5 0.32924 -4.4552
## 6 4.36840  9.6718
```

Now, a bit of exploratory data analysis:

```
attach(notes)
plot(V1, V2,
     col="yellow", pch=20,
     xlab="Property 1",
     ylab="Property 2")
```



The function `kmeans()` performs K-means clustering in `R`.

```
km.out <- kmeans(notes, 2, nstart = 20)
km.out
```

```
## K-means clustering with 2 clusters of sizes 783, 589
##
## Cluster means:
##           V1        V2
## 1  0.8664667  6.046704
## 2 -0.1415258 -3.560442
##
## Clustering vector:
##   [1] 1 1 2 1 2 1 1 2 1 1 1 2 1 1 1 2 1 1 2 1 1 2 1 1 1 2 2 1 1 2 2 1 1 1 2 1 1
##  [38] 2 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1 2 1 2 1 1 2 1 1 2 1 1 1 2 2 1 2
##  [75] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 1 1 1 2 1 2 1 2 1 1
##  [ reached 'max' / getOption("max.print") -- omitted 1272 entries ]
##
## Within cluster sum of squares by cluster:
## [1] 14250.46 12687.54
##  (between_SS / total_SS =  53.8 %)
##
## Available components:
##
## [1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
## [6] "betweenss"    "size"         "iter"         "ifault"
```

```
#here are the cluster assignments
km.out$cluster
```

```
##   [1] 1 1 2 1 2 1 1 2 1 1 1 2 1 1 1 2 1 1 2 1 1 2 1 1 1 2 2 1 1 2 2 1 1 1 2 1 1
##  [38] 2 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1 2 1 2 1 1 2 1 1 2 1 1 1 2 2 1 2
##  [75] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 1 1 1 2 1 2 1 2 1 1
##  [ reached 'max' / getOption("max.print") -- omitted 1272 entries ]
```

```
#here are the cluster centers
km.out$centers
```

```
##           V1        V2
## 1  0.8664667  6.046704
## 2 -0.1415258 -3.560442
```

It all works out much better visually.

```
#cloud of points with colors corresponding to cluster
plot(notes, col = (km.out$cluster + 1),
    main = "K-Means Clustering Results with K = 2",
    xlab = "Property 1", ylab = "Property 2", pch = 20, cex = 0.5)
#adding the centers
points(km.out$centers, col="blue", pch=20)
```


K−Means Clustering Results with K = 2
Property 2
10
5
0
−5
−10
−6
−4
−2
0
2
4
6
Property 1